Practice 2 E03
Aquaculture Schema Enforcement and Evolution

Submission Template

Level	Duration	Marks	Tasks	Screenshots
Easy	15 minutes	10	4	4

Trainee name: Soumiya Ayyappan Candidate ID: 10877547

Catalog: cat_ltm10877547 Date: _____

Notebook worksheet pipeline or dashboard location: _____

Business scenario

A fish farm receives a new stocking batch with an additional feed grade. First demonstrate schema enforcement, then deliberately evolve the same table to accept the new attribute.

Environment and object names

Use your existing assigned catalog and the fixed schema practice_2. Create that schema if it does not exist. Create only this question's new external volume at its assigned ADLS child path. Use the instructor-provided full URI, not an external-location root.

Item	Required value
Catalog	Your existing assigned catalog
Schema	practice_2
External volume	p2_e03_files
Relative ADLS path	practice_2/e03/files/ beneath your assigned trainee root
Input folder	/Volumes/<trainee_catalog>/practice_2/p2_e03_files/input/
Required object	p2_e03_farm_stock

The instructor provides a unique trainee root and complete question URI before timing. The ten question paths are separate. Volumes store source files; managed Delta tables and pipeline outputs use schema managed storage outside volume paths.

Keep supplied file and folder spelling exact. Select your catalog in each execution context, including pipeline destinations and dashboard datasets. Questions are independent; original implementations and required saved outputs must remain reviewable.

Dataset description

P2_E03_Aquaculture_Baseline.csv

CSV with a header row; 3 data rows and 6 columns. Blank cells mean missing values. Dates use YYYY-MM-DD; Boolean values use true or false.

Column	Required type	Description
stock_id	Text	Stock id
farm_id	Text	Farm id
species	Text	Species
fingerlings	Integer	Count stocked in the individual record
water_temp_c	Decimal 8 1	Water temp c
stock_date	Date YYYY-MM-DD	Stock date

P2_E03_Aquaculture_Incoming.csv

CSV with a header row; 2 data rows and 7 columns. Blank cells mean missing values. Dates use YYYY-MM-DD; Boolean values use true or false.

Column	Required type	Description
stock_id	Text	Stock id
farm_id	Text	Farm id
species	Text	Species
fingerlings	Integer	Count stocked in the individual record
water_temp_c	Decimal 8 1	Water temp c
stock_date	Date YYYY-MM-DD	Stock date
feed_grade	Text	Incoming-only field used for controlled schema evolution

Business rules and required columns

1. Start with automatic schema expansion disabled. Retain the original table throughout both append attempts.

2. First attempt the seven-column append without allowing evolution and capture its rejection. Verify unchanged state before proceeding.

3. Then permit additive expansion for that append and add the same two incoming records once. Do not remove feed_grade, overwrite the table or fill old feed grades.

Implementation flow

Prepare access → process data → compare expected outputs → capture evidence.

Task 1 Prepare governed source access

3 minutes | 2 marks | Exactly 1 screenshot

1. Use your existing assigned catalog. Create practice_2 if it does not exist and select it. Do not create a catalog.

2. Create a new EXTERNAL volume named p2_e03_files at the instructor-assigned full cloud URI for this question under the approved ADLS external location. Verify external type and the assigned storage location. Do not create a storage credential or external location.

3. Create an input folder in this volume and upload P2_E03_Aquaculture_Baseline.csv, P2_E03_Aquaculture_Incoming.csv. Preserve the exact filenames.

Expected output

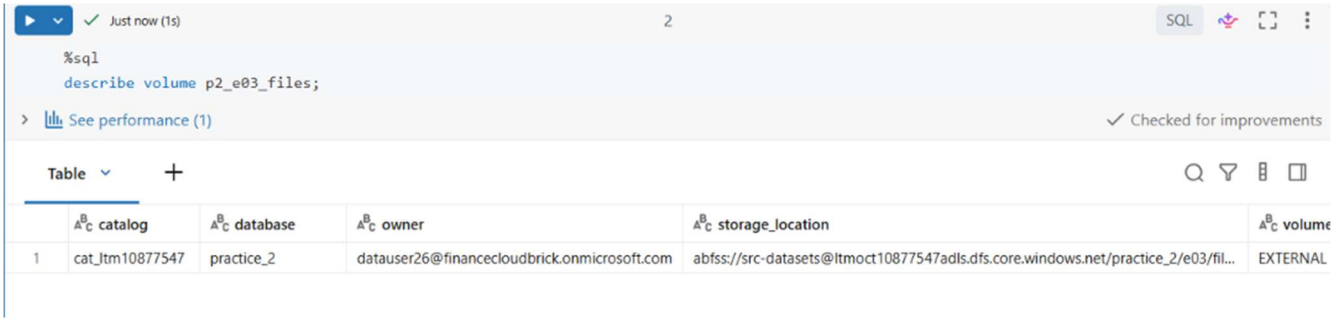
The question external volume, assigned URI and required input file or files are visible.

Required screenshot

External-volume details showing your catalog, practice_2, external type and assigned URI, with every required input filename visible.

Filename: P2_E03_T01.png. Place it in Task 1 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below



Evidence reference or workspace location:
/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/newseteasy3

Task 2 Create stocking baseline

3 minutes | 2 marks | Exactly 1 screenshot

1. Create managed Delta table p2_e03_farm_stock from the typed baseline.

2. Confirm six columns, three rows and total fingerlings, with schema expansion disabled.

Expected output

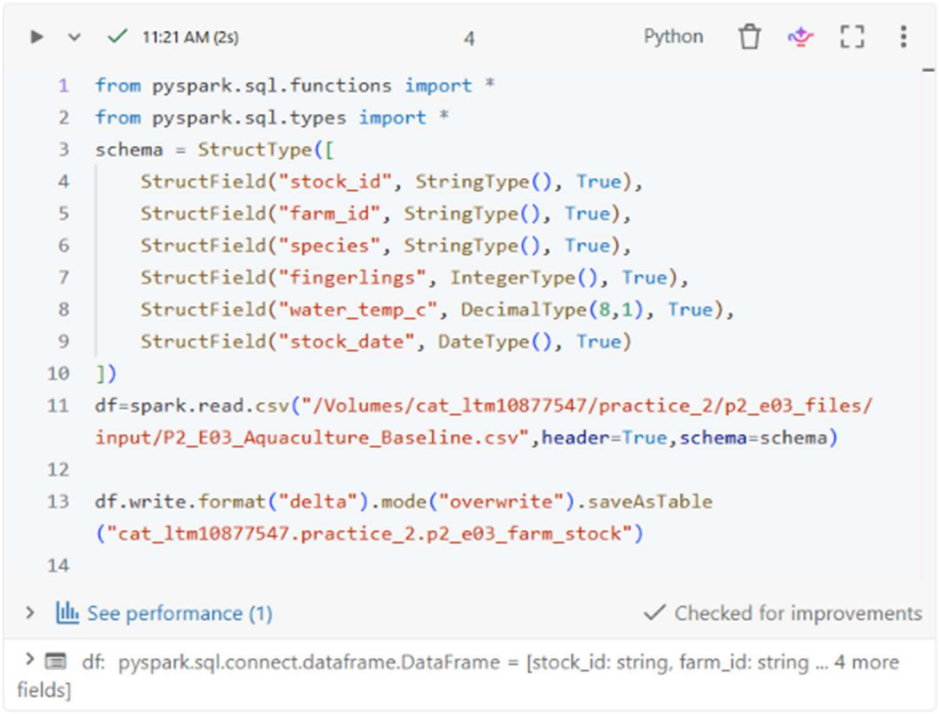
3 rows; 6 columns; 300 fingerlings; feed_grade absent.

Required screenshot

Your baseline implementation, schema and baseline row and fingerling totals.

Filename: P2_E03_T02.png. Place it in Task 2 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below

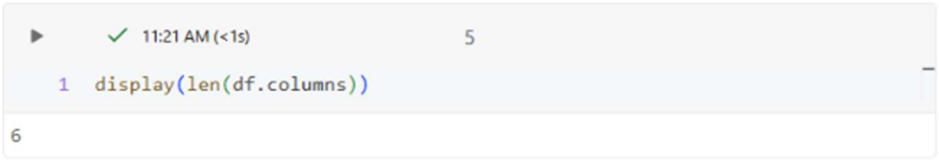


```
11:21 AM (2s)   4   Python

1  from pyspark.sql.functions import *
2  from pyspark.sql.types import *
3  schema = StructType([
4      StructField("stock_id", StringType(), True),
5      StructField("farm_id", StringType(), True),
6      StructField("species", StringType(), True),
7      StructField("fingerlings", IntegerType(), True),
8      StructField("water_temp_c", DecimalType(8,1), True),
9      StructField("stock_date", DateType(), True)
10 ])
11 df=spark.read.csv("/Volumes/cat_ltm10877547/practice_2/p2_e03_files/
   input/P2_E03_Aquaculture_Baseline.csv",header=True,schema=schema)
12
13 df.write.format("delta").mode("overwrite").saveAsTable
   ("cat_ltm10877547.practice_2.p2_e03_farm_stock")
14

> See performance (1)                    ✓ Checked for improvements

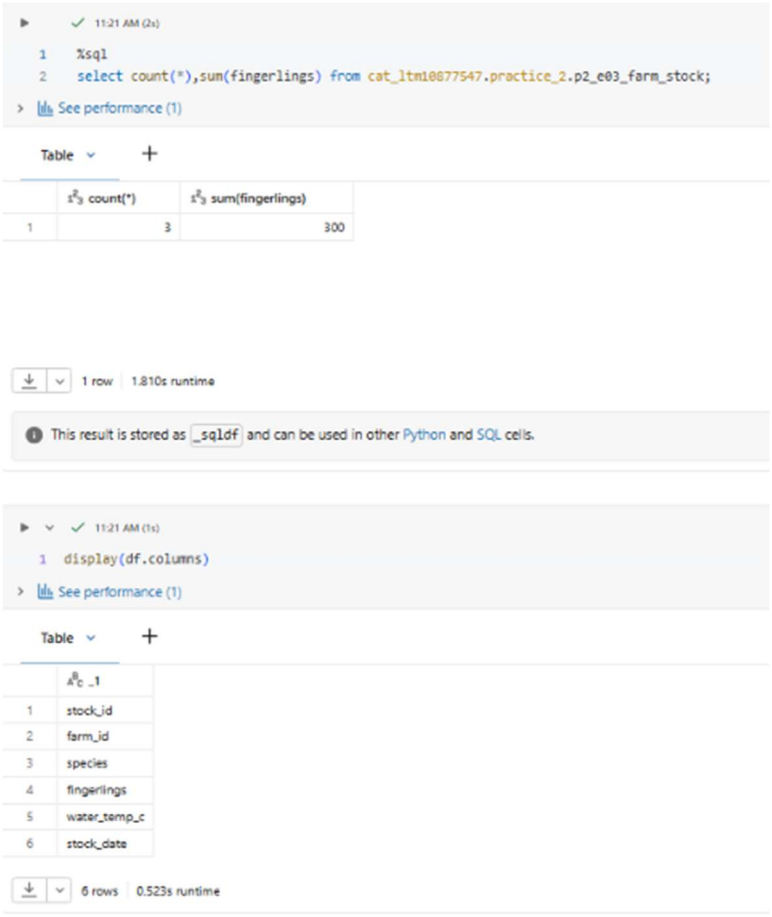
> df: pyspark.sql.connect.dataframe.DataFrame = [stock_id: string, farm_id: string ... 4 more
fields]
```



```
11:21 AM (<1s)   5

1  display(len(df.columns))

6
```



```
%sql
select count(*),sum(fingerlings) from cat_ltm10877547.practice_2.p2_e03_farm_stock;
```

See performance (1)

Table

	count(*)	sum(fingerlings)
1	3	300

1 row | 1.810s runtime

This result is stored as `_sqldf` and can be used in other Python and SQL cells.


```
display(df.columns)
```

See performance (1)

Table

	_1
1	stock_id
2	farm_id
3	species
4	fingerlings
5	water_temp_c
6	stock_date

6 rows | 0.523s runtime

Evidence reference or workspace location: _____

Task 3 Demonstrate rejected schema change

5 minutes | 3 marks | Exactly 1 screenshot

1. Read the incoming seven-column batch and attempt append without allowing expansion.

2. Retain the actual mismatch error. Immediately verify the target remains at its original schema and totals.

Expected output

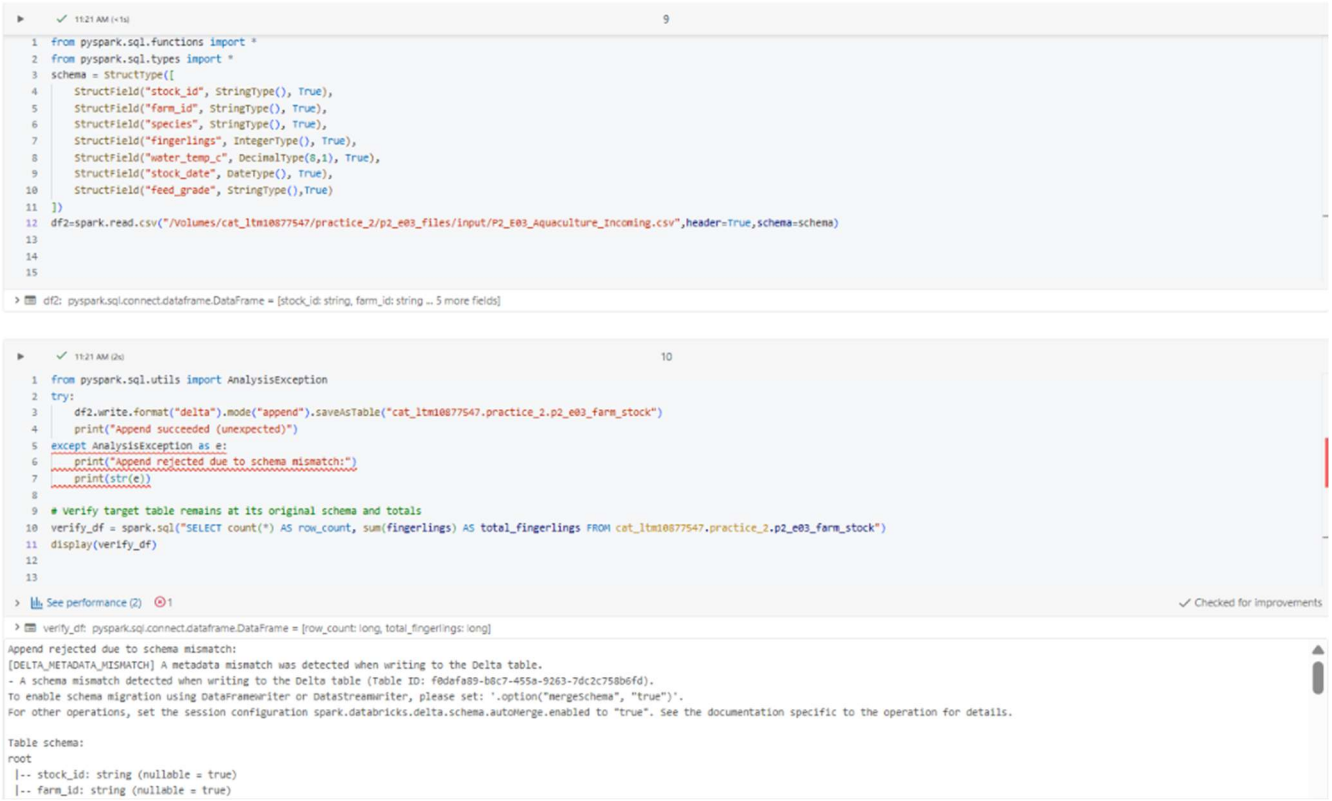
Append rejected; 3 rows and300 fingerlings remain; feed_grade absent.

Required screenshot

Your incompatible append attempt, readable error and unchanged-state checks.

Filename: P2_E03_T03.png. Place it in Task 3 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

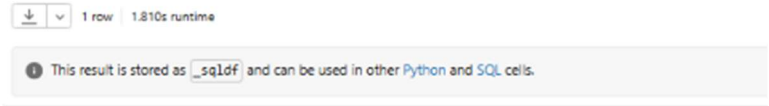
Paste the one required screenshot below



```
11:21 AM (<1s)                                                
```







Evidence reference or workspace location: _____

Task 4 Evolve the schema and reconcile

4 minutes | 3 marks | Exactly 1 screenshot

1. Allow the additive feed_grade column and append the two new records exactly once to the same table.

2. Display all final rows and schema; verify new grades, missing old grades, uniqueness and total fingerlings.

Expected output

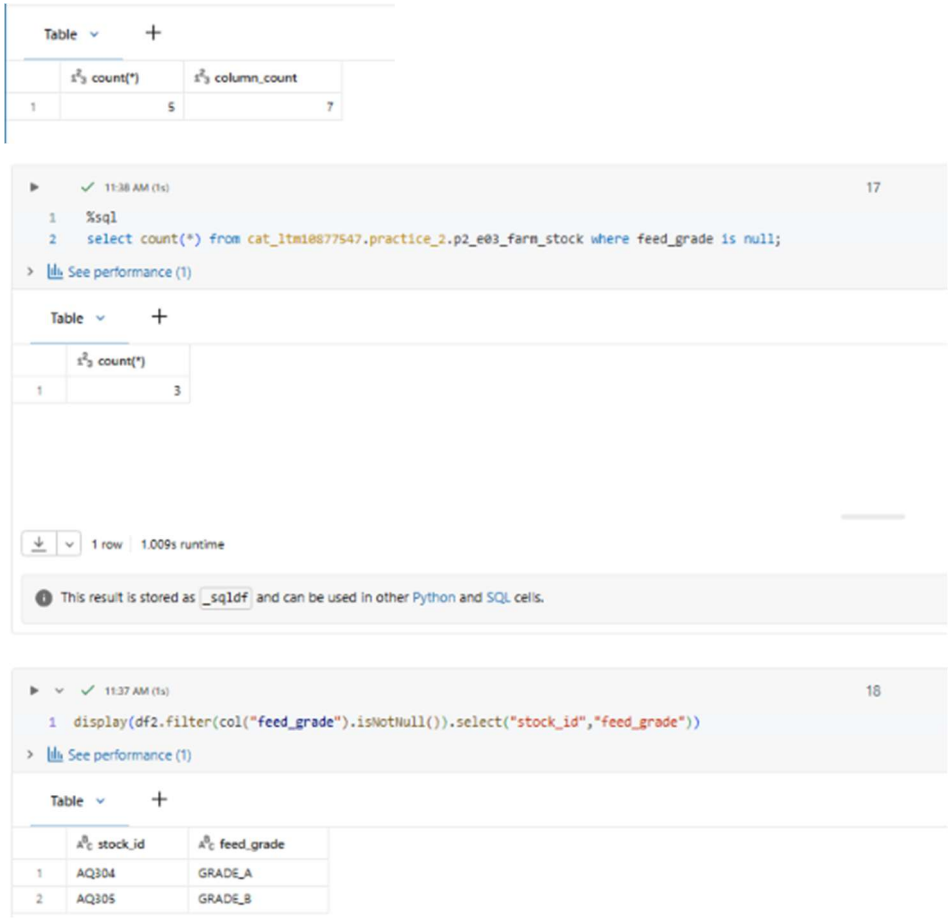
5 unique rows; 7 columns; 400 fingerlings; old3 feed grades missing; AQ304: GRADE_A and AQ305: GRADE_B.

Required screenshot

Your successful additive append implementation and final schema, records and reconciliation.

Filename: P2_E03_T04.png. Place it in Task 4 of the matching submission template. Your implementation and output must be readable together where requested. Related panels may be captured in one image; tiny text or unreadable stitched images are not acceptable.

Paste the one required screenshot below



Table

	count(*)	column_count
1	5	7

11:38 AM (1s) 17

```
%sql
select count(*) from cat_ltm10877547.practice_2.p2_e03_farm_stock where feed_grade is null;
```

See performance (1)

Table

	count(*)
1	3

1 row | 1.009s runtime

This result is stored as _sqldf and can be used in other Python and SQL cells.

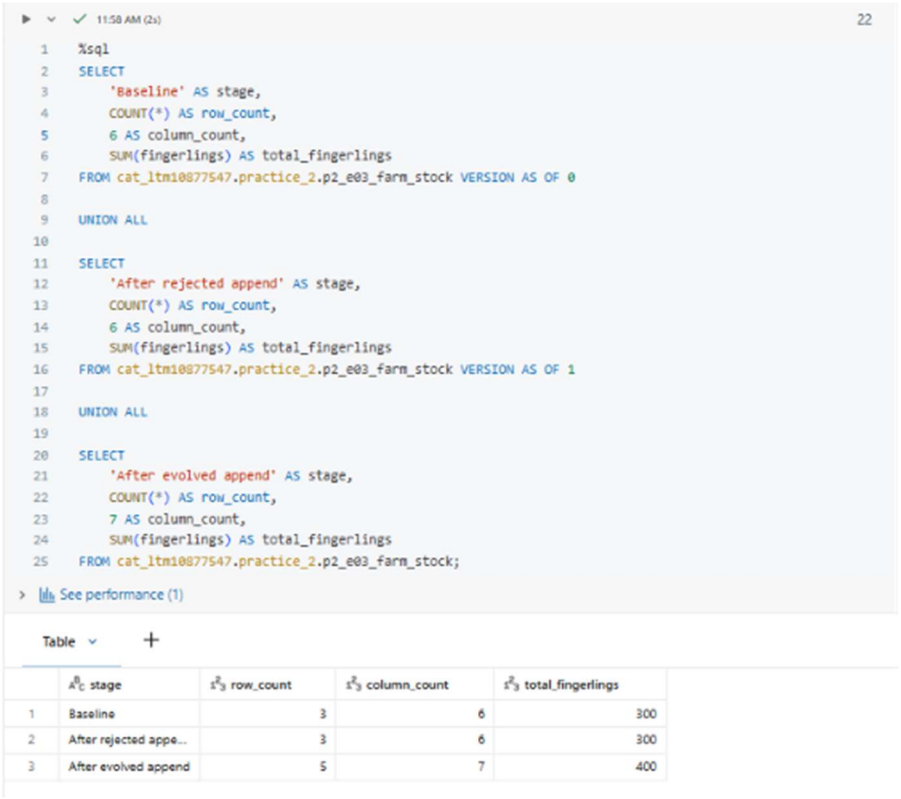
11:37 AM (1s) 16

```
display(df2.filter(col("feed_grade").isNotNull()).select("stock_id","feed_grade"))
```

See performance (1)

Table

	stock_id	feed_grade
1	AQ304	GRADE_A
2	AQ305	GRADE_B



```
%sql
SELECT
    'Baseline' AS stage,
    COUNT(*) AS row_count,
    6 AS column_count,
    SUM(fingerlings) AS total_fingerlings
FROM cat_ltm10877547.practice_2.p2_e03_farm_stock VERSION AS OF 0

UNION ALL

SELECT
    'After rejected append' AS stage,
    COUNT(*) AS row_count,
    6 AS column_count,
    SUM(fingerlings) AS total_fingerlings
FROM cat_ltm10877547.practice_2.p2_e03_farm_stock VERSION AS OF 1

UNION ALL

SELECT
    'After evolved append' AS stage,
    COUNT(*) AS row_count,
    7 AS column_count,
    SUM(fingerlings) AS total_fingerlings
FROM cat_ltm10877547.practice_2.p2_e03_farm_stock;
```


	stage	row_count	column_count	total_fingerlings
1	Baseline	3	6	300
2	After rejected appe...	3	6	300
3	After evolved append	5	7	400

Evidence reference or workspace location: _____

Expected final output

Stage	Rows	Columns	Fingerlings
Baseline	3	6	300
After rejected append	3	6	300
After evolved append	5	7	400

Evaluation criteria and screenshot map

Task	Criterion	Marks
1	Correct catalog schema external volume and URI	1
1	Correct input folder and all source files	1
2	Correct baseline table and disabled expansion	1

Task	Criterion	Marks
2	Correct schema and totals	1
3	Actual incompatible append rejected	2
3	Original schema rows and totals verified before retry	1
4	Correct additive append without overwrite or duplicate input	1
4	Correct old and new grades and schema	1
4	Final count uniqueness and fingerling total correct	1

Task	Time minutes	Marks	Screenshots	Filename
1	3	2	1	P2_E03_T01.png
2	3	2	1	P2_E03_T02.png
3	5	3	1	P2_E03_T03.png
4	4	3	1	P2_E03_T04.png

Total: 15 minutes, 10 marks and exactly 4 screenshots. No extra screenshots are required.

Marks are awarded for the stated implementation and observable task outcomes. A screenshot supports the criterion assigned to its task; it does not create an additional criterion. Retain original work so that implementation and saved outputs remain reviewable.

Submission requirements

Submit the matching template with exactly 4 readable task screenshots. Retain your original implementation and required objects in your trainee workspace folder and add their names or links to the template. Confirm expected outputs and fixed names. Preserve table history and change evidence. No additional report or validator is required.